WildTrack Conservation Alliance

Database Practice Project

The Scenario

WildTrack Conservation Alliance is a multi-country wildlife conservation programme that protects endangered species across a network of forest reserves and national parks. Field rangers fit GPS tracking collars onto individual animals — tigers, elephants, snow leopards, river dolphins, and dozens of other threatened species — and each collar quietly beams back its location, speed, temperature, and battery health every few hours, day and night.

Over the past few years, this has grown into a live operational database that rangers, biologists, the data team, and donor relations all depend on — to plan patrols, catch failing equipment before an animal goes dark, spot signs of poaching or distress, and report back to the funders keeping the whole programme running. The database sits at the centre of all of it.

The database is built around four connected tables:

- Reserves — every forest reserve or national park in the network
- Species — every species currently being monitored, along with its conservation status
- Animals — every individual animal fitted with a collar, linked to its species and its home reserve
- GPSLogs — every single location ping sent back by a collar, linked to the animal wearing it

Altogether the database holds 10,000 records across these four tables — the vast majority of them GPS pings, since a handful of collars quietly generate an enormous stream of location data every day.

Below are 75 real situations that come up in running a programme like this. Read each one and work out how you'd solve it using the database — nothing is labelled, so the first step is always deciding what the situation is actually asking you to do.

Cases

1. A researcher's fieldwork contract has just ended, and their access to the database needs to be fully withdrawn with immediate effect.

2. Group all GPS pings by the animal they belong to, and flag animals with fewer than 5 total pings recorded so far — their collars might be malfunctioning.

3. Some reserves may not have a single animal collared yet. Identify exactly which ones, since this is a coverage gap worth flagging to the programme director.

4. The steering committee wants to know, in one number, how many distinct species are currently being monitored across all reserves combined.

5. An intern wants to flag any reserve name that's unusually long (more than 25 characters) so it can be shortened for the mobile app's limited screen space.

6. Sort each animal's estimated age into 'Juvenile', 'Adult', or 'Senior' bands using sensible thresholds, to help the biology team study age distribution across the population.

7. For each reserve, smooth out momentary sensor glitches by calculating an average battery level using only the current reading and the two immediately before it.

8. Break this into two steps: first find each animal's most recent log date, then use that to identify which animals haven't reported anything in the last 10 days.

9. Some tag codes were typed with inconsistent capitalization by different rangers over the years. Produce a clean, uniformly capitalized list for the tracking dashboard.

10. First tally each species' total tracked population, then bring in reserve details to see which reserves are hosting the heaviest concentrations of the most vulnerable species.

11. Identify every animal whose collar was fitted more than two years ago — these units are approaching the end of their expected working life and should be flagged for replacement.

12. Find the lowest battery level ever recorded across all collars — that particular device may need an urgent field swap.

13. A ranger wants to know which day of the week sees the most recorded animal movement, purely to plan patrol shifts more efficiently.

14. The dashboard team wants something they can plug straight into their reporting tool that always reflects each reserve's current animal count and average battery health, without re-deriving it manually each time.

15. Pull every GPS log recorded during the month of June this year, to study how the tracked animals behaved during the onset of monsoon rains.

16. For each species, build a running count of how many animals have been collared over time, ordered by fitting date, to visualize how the programme has grown.

17. Prepare a field report listing every animal along with its species name and the reserve it currently roams in, all on a single line per animal.

18. The system currently lets someone log a battery level of 150% by mistake. What change would permanently make this impossible going forward?

19. For a specific animal that's causing concern, work out how many hours have passed between its very first logged position and its most recent one, to check whether the collar is still reporting normally.

20. For each animal, work out how many days have passed since its collar was first fitted — this determines which ones are due for the routine 180-day battery check.

21. Flag each reserve as 'Overcrowded', 'Balanced', or 'Under-monitored' based on how many animals are currently tracked there relative to the reserve's area.

22. For the upcoming funding proposal, work out the average estimated age of every Critically Endangered animal currently in the system.

23. Label each collar's current battery level as 'Critical' (below 15%), 'Low' (15–40%), or 'Healthy' (above 40%), so field teams instantly know which ones need urgent attention.

24. Identify which reserves have recorded an average animal speed above 15 km/h over the last month, which could point to unusual stress or fleeing behaviour spreading across the local population.

25. For each species, count the number of individuals currently being tracked, and highlight any species where fewer than 10 individuals are monitored — a conservation priority signal.

26. For each animal, flag any of its own GPS logs where the recorded speed is noticeably higher than that same animal's own typical speed — a possible burst of unusual activity worth a closer look.

27. Track down which reserve is home to the animal that recorded the single highest speed anywhere in the entire dataset.

28. Rangers constantly search GPS logs using a specific animal's ID to pull up its full movement history instantly, but this lookup has started to feel sluggish now that the log table has grown into the thousands. What would you do about it?

29. An animal previously assumed dead has just been confirmed alive and re-collared in the field. Reflect this by recording its new collar-fitting date.

30. For a laminated field handout, combine each animal's tag code and species name into a single readable label, separated by a dash.

31. Work out how many animals have been fitted with collars in each individual reserve, to see where the tracking effort is concentrated.

32. Work this out in two stages: first tally up each animal's total number of logs this month, then use that to shortlist animals logged fewer than 3 times — these may have gone quiet.

33. Estimate how many field battery-replacement trips are needed this month by finding the average battery level currently across all collars.

34. For every species being tracked, list out all the reserves where at least one individual of that species currently carries a collar.

35. List all animals currently based in one of the three largest reserves by area.

36. Identify any animal that has never had a single GPS ping recorded against it — this is a critical operational failure that needs immediate attention.

37. The programme director wants a single number for the board meeting: exactly how many animals are currently being tracked across the entire alliance.

38. Going forward, the programme wants to record which manufacturer supplied each animal's GPS collar. Adjust the animal records so this detail can be captured.

39. Two rangers in different reserves accidentally try to register the same tag code for two different animals on the same day. What should stop this from happening?

40. For every reserve, surface the single fastest animal recorded there without losing any of the reserve's other animal rows in the process — every animal should still show up in the result.

41. Find the single highest speed ever recorded by any collar so far — this reading could point to a chase, an escape, or a predator encounter worth investigating.

42. Count the animals collared in each reserve, and shortlist only the reserves currently tracking more than 15 animals — these may need additional ranger support.

43. Group all logs by reserve and identify which reserves have an average collar battery level below 20% — these need a battery-replacement drive scheduled immediately.

44. The annual conservation report needs a single consolidated result: for every reserve, the number of animals it holds, their average age, the average battery health of their collars, and a clear flag for any reserve whose average battery health has dropped below an acceptable level.

45. A recent firmware change altered how battery percentage gets reported by one collar model. Correct all currently stored battery readings for the affected reserve so they reflect the new calibration.

46. Produce a single combined list of every reserve and every animal together — including reserves with zero animals and any animal record that's somehow missing its reserve — so nothing gets left out of the audit.

47. Find every species that has at least one tracked individual living in a reserve that was established before the year 2000.

48. After a recent data cleanup, a few GPS logs may now point to an animal ID that no longer exists in the system. Track down these orphaned records.

49. Calculate a combined activity score per animal using its log count and average speed as an intermediate result, then use that to rank reserves by overall animal activity.

50. A batch of very old GPS logs from a decommissioned first-generation collar model is cluttering the database and is no longer needed for any analysis. Remove exactly those records and nothing else.

51. A junior data-entry volunteer tries to save a new animal record without specifying which species it belongs to. What should prevent that incomplete record from being saved?

52. Build an intermediate reserve-by-reserve average battery level, then use it to flag every reserve sitting below the alliance-wide average.

53. The reporting dashboard constantly filters by reserve and date range together, and the two filters are almost always used side by side. How would you set the database up so this specific combined search runs faster?

54. A bulk import left several reserve names with accidental leading or trailing spaces. Clean these up before they go into the annual printed report.

55. The rangers' mobile app needs a simplified, always-ready snapshot showing each animal's tag code, species, reserve, and latest battery level, without the field team having to reconstruct the full picture from scratch every single time.

56. The donor relations team keeps asking for the same list — every Critically Endangered animal with its reserve and species details. Set up something they can pull directly, without you explaining the underlying tables every time they ask.

57. The data team wants to understand overall data volume: how many GPS pings have been logged in total since the programme began?

58. A field officer accidentally types an animal's estimated age as a negative number. How do you make sure this kind of entry is rejected the moment it's submitted?

59. Split all tracked animals into three roughly equal age tiers, to help the biology team design three different age-specific conservation strategies.

60. Pull every GPS log belonging specifically to the one animal that recorded the lowest battery level ever logged in the system.

61. A data-entry mistake caused an entire reserve's animals to be logged under the wrong reserve altogether. Correct this in one sweep rather than animal by animal.

62. For each animal, show what the very next recorded speed reading will be, placed right alongside the current one, to help spot a pattern building up across consecutive readings.

63. Within each reserve, order its animals by estimated age, oldest first, to identify individuals that might need closer health monitoring.

64. A new GPS ping arrives tagged to an animal ID that was never registered in the system. Explain what should happen to this record, and why it must not be allowed to slip through quietly.

65. Cross-check reserves against recent activity to identify any reserve where absolutely no GPS pings have come in during the past seven days — this could mean a communication blackout or a bigger problem on the ground.

66. Across the entire programme, order every GPS log by recorded speed to surface the season's most extreme movement events, making sure that if two logs share the exact same top speed, neither one gets unfairly skipped in the ordering.

67. Group the collaring activity by the exact quarter of the year each animal was fitted, to compare how collaring effort has changed across different phases of the programme.

68. For each animal, looking at its logs in date order, show what its battery level was on the reading immediately before the current one — this helps catch any sudden, unexplained battery drop.

69. A new field researcher is joining the team and should only be able to look at animal and GPS data, never change any of it. Set up their access accordingly.

70. List every animal along with its most recent recorded coordinates, making sure animals whose collars haven't reported anything yet still appear on the list rather than disappearing.

71. The lead veterinarian needs the ability to update animal health-related details directly, unlike the rest of the field staff. Set up an access level that reflects this difference.

72. Identify every animal that belongs to whichever single species currently has the highest number of tracked individuals.

73. After several months and a few team changes, nobody quite remembers what was set up to speed up lookups on the GPS log table. Check what's currently in place.

74. Based on recorded speed, tag each GPS log as 'Resting', 'Moving', or 'Fast Movement', to help spot unusual chase or escape events buried in the raw data.

75. For a nocturnal-behaviour study, pull all GPS logs that were recorded between 6 PM and 6 AM.